CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Theory Examination

May 2014

ME 101.01 Engineering Graphics

Date: 26.05.2014, Monday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Figures drawn in the question paper are not to the scale.
6. Dimensions shown in the figure are in mm.
7. Show all the necessary dimensions in the solution.
8. Figures to right indicate the full marks.
9. Don't rub Construction/Projection lines after completion of drawing.

SECTION - I

Q.1 (a) Choose appropriate answer by rewriting the question. **[07]**

1. The isometric view of a sphere is always
   (a) a circle (b) an ellipse (c) a parabola (d) a semicircle
2. Give the full form of BIS for engineering drawing.
   (a) British Information Service
   (b) Bureau of Indian Standards
   (c) Bureau of International Standards
   (d) Business Information System
3. A polygon having 30 sides, what will be the angle between two adjacent sides.
   (a) 12 (b) 120 (c) 168 (d) None of these
4. A circular plate is inclined to 57^0 with HP and 33^0 with VP, what will be the angle of plate with profile plane.
   (a) 0^0 (b) 45^0 (c) 90^0 (d) None of these
5. If a development of cone is a semicircle with a radius R then what is the value of base diameter.
   (a) R (b) R/2 (c) 2R (d) None of these
6. In isometric projection view, the true length is drawn at which angle.
   (a) 30^0 (b) 45^0 (c) 60^0 (d) None of these
7. It is visible a rectangle in front view and top view, which of the solid is possible.
   (a) hexagonal prism (b) cube (c) cylinder (d) None of these

Q.1 (b) Answer the following questions. **[10]**

1. Give the application of AutoCAD. **[02]**
2. Define R.F. Construct a scale of 1:4 to show centimeters and long enough to measure upto 5 decimeters. Show a length of 3.7 decimeter on it. **[05]**
3. A point *P* is 30 mm above H.P. & 30 mm in front of V.P. Draw the projection of point *P*. Mention the possible outcome quadrant. **[03]**

OR

Q.1 (b) The major axis and the minor axis of an ellipse are 120 mm and 70 mm long. Draw the ellipse by concentric circle method. Draw the normal and tangent at any point on the curve. Also find out the eccentricity of an ellipse. [1

Q. 2 A fan is hanging in the center of the room of 4m X 3m X 3m height. The center of the fan is 0.75 m below the ceiling. The switch of this fan is on 3m X 3m size wall at the center height and 0.5 m from the adjacent wall. Find the distance between the fan center and the switch. [08]

OR

Q. 2 A point *P* is 30 mm above HP and is in first quadrant. Its shortest distance from XY line is 60 mm. Draw its plan and elevation. [08]

Q.3 Fig. 1 shows pictorial view of an object. Draw the following views using first angle projection system. [10]
(1) Front view in the direction 'X' and (2) Top view,

SECTION - II

Q.4 A 30° – 60° set square of longest side 100 mm long is in VP and 30° inclined to HP while its surface is 45° inclined to VP. Draw its projections. [07]

OR

Q.4 *PQRS* is a rhombus of diagonal *PR* is 60 mm and *QS* is 40 mm the corner *P* is in the HP and the plane is inclined to the HP such that the plan appear is square. The plan of diagonal *PR* makes an angle of 20° to the VP. [07]

Q. 5 (a) A cone of base diameter 60 mm and height of 75 mm is resting on HP on its base in such a way that axis is inclined at 30° to HP and plan of axis is inclined at 30° to VP. Draw projection of the cone when apex is nearer to VP. [09]

Q. 5 (b) A cylinder, diameter of base 60 mm and height 80 mm is resting on HP on its base. It is cut by AIP, which makes an angle 30° to HP and bisecting the axis. Draw the development of cut cylinder. [09]

OR

Q. 5 (a) A tetrahedron of 30 mm side is resting with one of its edges on H.P. The edge on which it rests is inclined at 45° to VP and a face containing that edge is inclined 30° to HP. Draw the projections of solid. [09]

Q. 5 (b) A pentagonal pyramid (40 x 70) is resting on HP on its base with one of the edges of base away from VP is parallel to VP. It is cut by two AIPs No. 1 and No. 2 both inclined at 45° to the HP passing through points 30 mm and 35 mm from apex on axis respectively. Draw the development of the cut pyramid. Show also effect in plan. [09]

Q.6 Fig. 2 shows the orthographic projections of an object in first angle projection system. Draw its isometric view. [10]

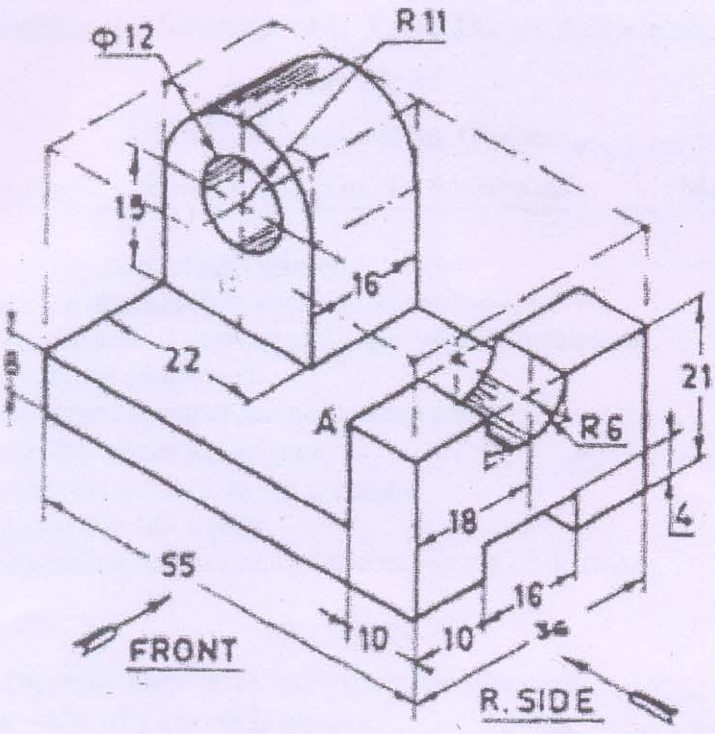

Figure 1

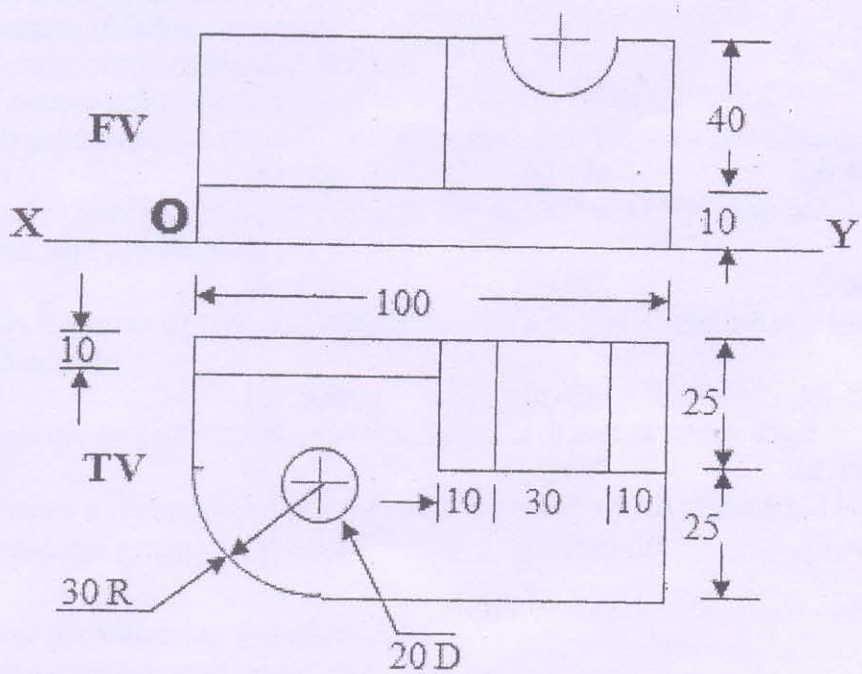

Figure 2